

6G WIRELESS TECHNOLOGY

Seminar Report

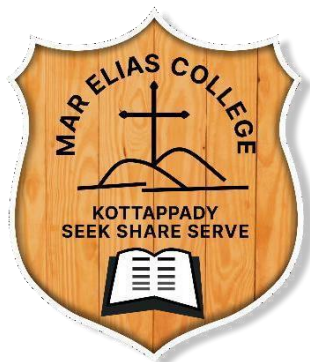
Submitted By

ARUN P S: 230021080343

In partial fulfillment of the requirements for award of the degree of

Bachelor of Science in Computer Science

MAHATMA GANDHI UNIVERSITY, KOTTAYAM – 686560



MAR ELIAS COLLEGE, KOTTAPPADY P. O. – 686695

(Affiliated to Mahatma Gandhi University, Kottayam 686560) Ac-credited by NAAC with B++ Grade

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CERTIFICATE

This is to certify that the seminar report titled **6G WIRLESS TECHNOLOGY** sub-mitted by **ARUN P S (230021080343)** towards partial fulfilment of the requirement for the award of the degree of Bachelor of Science in Com- puter Science is a report of bonafide work carried out by him during the academic year 2025-2026.

Ms. Sreekala c
Seminar In-
charge

Ms. Poornima N.
HOD

Dr. Joseph T Moolayil
Principal

DECLARATION

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I, hereby declare that this seminar report entitled **6G WIRLESS TECHNOLOGY** has been written and submitted to the department of Computer Science and Applications, Mar Elias College, Kottappady in partial fulfillment of the award of the degree of Bachelor of Science in Computer Science is an authentic record of my original work. The empirical findings in the report are on information collected by me and are not copied from anywhere.

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ACKNOWLEDGMENT

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6G WIRELESS TECHNOLOGY

Abstract—

6G Wireless Technology represents the next generation of mobile communication systems, expected to succeed 5G by the year 2030. It aims to deliver ultra-high data rates, extremely low latency, massive connectivity, and intelligent communication systems powered by artificial intelligence. 6G is envisioned to support advanced applications such as holographic communication, extended reality (XR), autonomous systems, smart cities, and digital twins.

This seminar report discusses the fundamental concepts of 6G wireless technology, its architecture, key features, applications, advantages, and limitations. It also highlights how 6G will revolutionize communication by integrating AI, terahertz frequencies, and satellite-based networks to enable seamless global connectivity.

Keywords: 6G, Wireless Communication, Terahertz, Artificial Intelligence, Smart Networks, Future Technology

INDRODUCTION

Wireless communication has evolved rapidly from 1G to 5G, transforming the way people communicate and access information. While 5G offers high-speed and low-latency communication, emerging technologies and applications demand even higher performance, intelligence, and reliability. This has led to the development of 6G Wireless Technology.

6G is expected to provide data rates up to terabits per second (Tbps), near-zero latency, and ubiquitous connectivity. It will integrate communication, computing, sensing, and artificial intelligence into a unified platform. The goal of 6G is not only to improve speed but also to create intelligent, adaptive, and context-aware networks.

WHAT IS 6G WIRELESS TECHNOLOGY?

6G Wireless Technology is the sixth generation of mobile communication systems that will operate beyond 5G capabilities. It is designed to use higher frequency bands such as terahertz (THz) and integrate AI-driven network management to support next-generation applications.

Unlike previous generations, 6G focuses on human-centric communication, enabling immersive digital experiences and real-time interaction between humans, machines, and environments.

WHY DO WE NEED DIGITAL TWIN?

01. Higher Data Rates-

To support holographic communication and XR applications.

02. Ultra-Low Latency-

Required for autonomous vehicles and remote surgeries.

03. Massive Connectivity –

To connect billions of IoT and smart devices

04. Global Coverage-

Integration of satellite and aerial networks.

05. Intelligent Networks –

AI-based. optimization and self-healing Networks.

CHARACTERISTICS OF 6G

1. **Terahertz Communication** – Utilizes THz spectrum for ultra-fast data transmission.
2. **Artificial Intelligence Integration** – AI for network optimization and decision-making.
3. **Ultra-Low Latency** – Latency less than 1 millisecond.
4. **High Reliability** – Supports mission-critical applications.
5. **Energy Efficiency** – Reduced power consumption for sustainable communication.
6. **Sensing and Communication Integration** – Enables environmental sensing along with data transfer.

ARCHITECTURE OF 6G

The architecture of 6G consists of the following layers :

Device Layer • Device Layer:

Consists of smart devices, sensors, and IoT nodes that generate and collect data. These devices communicate with the network and support applications such as smart homes, healthcare monitoring, and autonomous systems.

• Access Network Layer:

Includes advanced base stations that use terahertz (THz) frequency bands to provide ultra-high-speed and low-latency communication between devices and the core network.

• AI Layer:

Uses artificial intelligence for intelligent network control and optimization. Helps in traffic management, resource allocation, fault detection, and self-healing of the network.

• Core Network Layer:

Responsible for high-speed data routing, processing, and storage. Ensures reliable communication, quality of service, and seamless data flow across the network.

• Satellite and Aerial Layer:

Provides global connectivity using satellites, high-altitude platforms, and unmanned aerial vehicles (UAVs). Ensures network coverage in remote, rural, and disaster-prone areas

FEATURES OF 6G

• Ultra-High Data Rates (Up to Tbps):

Supports extremely fast data transmission, enabling instant downloads and smooth handling of data-intensive applications like holographic communication and high-resolution streaming.

• Near-Zero Latency Communication:

Provides almost real-time data transfer with minimal delay, which is essential for applications such as autonomous vehicles, remote surgery, and real-time gaming.

• AI-Driven Network Management:

Uses artificial intelligence to monitor, control, and optimize network operations automatically, improving efficiency, reliability, and overall performance.

• Holographic and XR Support:

Enables advanced immersive experiences including holographic communication and Extended Reality (AR/VR/MR) by providing high speed and low latency connectivity.

• Seamless Global Connectivity:

Integrates terrestrial networks with satellites and aerial platforms to ensure uninterrupted connectivity across urban, rural, and remote areas.

• Enhanced Security and Privacy:

Implements advanced encryption and intelligent security mechanisms to protect user data and ensure safe communication.

• Smart and Self-Adaptive Networks:

Allows the network to learn from real-time data and automatically adjust to changing conditions, ensuring optimal performance without manual intervention.

APPLICATIONS OF 6G

Holographic Communication:

Enables real-time 3D hologram-based communication, allowing people to interact as lifelike virtual images. It is useful for virtual meetings, education, medical consultations, and remote collaboration with ultra-low latency and high data rates.

Smart Cities and Smart Infrastructure:

Supports intelligent management of urban infrastructure such as traffic control, street lighting, waste management, and energy distribution. Helps improve sustainability, reduce congestion, and enhance quality of life through real-time data analysis.

Autonomous Vehicles and Drones:

Provides ultra-reliable and low-latency communication for self-driving cars and drones. Enables real-time navigation, collision avoidance, traffic coordination, and safe autonomous operations.

Remote Healthcare and Telesurgery:

Allows doctors to monitor patients remotely and perform surgeries using robotic systems. Ultra-low latency and high reliability ensure precise control, real-time feedback, and improved healthcare access in remote areas.

Extended Reality (AR/VR/MR):

Supports immersive experiences using Augmented Reality, Virtual Reality, and Mixed Reality. Used in education, training, gaming, virtual tourism, and industrial simulations with high-quality visuals and real-time interaction.

Industrial Automation:

Enables smart factories with real-time monitoring and control of machines. Improves production efficiency, reduces downtime, and supports predictive maintenance through intelligent and connected systems.

- **Digital Twins and Smart Manufacturing:**

Creates virtual replicas of machines, processes, and factories for real-time monitoring and optimization. Helps in improving productivity, reducing costs, and enabling data-driven decision-making in manufacturing industries.

ADVANTAGES OF 6G

- **Extremely High-Speed Communication:**

Provides ultra-fast data transmission with speeds up to terabits per second (Tbps). Enables seamless streaming, instant data transfer, and supports data-heavy applications like holographic communication and extended reality.

- **Improved Quality of Service:**

Ensures reliable, stable, and low-latency communication even in high-traffic conditions. Offers better user experience with reduced delays, fewer interruptions, and higher network availability.

- **Intelligent and Adaptive Networks:**

Uses artificial intelligence and machine learning to automatically optimize network performance. Enables self-learning, self-healing, and efficient resource management based on real-time conditions.

- **Supports Futuristic Applications:**

Facilitates advanced technologies such as autonomous vehicles, smart cities, remote surgeries, and digital twins. Makes future innovations possible with high reliability and ultra-low latency.

- **Enhanced Global Connectivity:**

Integrates terrestrial networks with satellites and aerial platforms to provide seamless connectivity worldwide. Ensures network access even in remote and rural areas.

DISADVANTAGES OF 6G

- **High Implementation Cost:**

Requires heavy investment in advanced hardware, new network infrastructure, satellites, and research development, making deployment expensive in the initial stages.

- **Complex Infrastructure Requirements:**

Needs highly sophisticated network architecture involving terahertz communication, AI-based systems, and satellite integration, which increases design and maintenance complexity.

- **Security and Privacy Concerns:**

Handling massive amounts of data and intelligent networks increases

the risk of cyberattacks, data breaches, and privacy violations if not properly secured.

- **Technological Challenges in THz Communication:**

Terahertz signals have limited range and are easily affected by obstacles like walls and weather conditions, making signal propagation and stability challenging.

- **Need for Advanced Hardware and Skills:**

Requires specialized devices, sensors, and skilled professionals with expertise in AI, networking, and advanced communication technologies.

CONCLUSION

6G Wireless Technology is expected to redefine the future of communication by providing intelligent, ultra-fast, and reliable connectivity. By integrating artificial intelligence, terahertz communication, and satellite networks, 6G will enable revolutionary applications that were previously unimaginable. Although there are several challenges in implementation, continuous research and development will make 6G a key enabler of future digital societies.

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